



亞洲大學
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Midterm Project Report

Advanced Computer Programming

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Chapter 1 Introduction

1.1 Github

- 1) **Personal Github Account:** <https://github.com/wangyalal>
- 2) **Group Project Repository:** <https://github.com/wangyalal/Group1.git>

1.2 Overview

This project is on extraction of repository metadata from a personal GitHub profile. It primarily utilizes Scrapy, a Python open-source web-crawling framework. To ensure high data integrity for the XML file, the following advanced features were implemented.

- **Item Loaders and Processors:** used MapCompose and TakeFirst to remove whitespace and ensure consistent formatting.
- **Data Fallback:** Implemented priority logic to ensure no null entries regarding the repository description (About).
- **Asynchronous URL Joining:** Used `response.urljoin()` to dynamically convert relative paths into absolute clickable URLs.

Chapter 2 Implementation

2.1 Class: RepoItem

This serves as the primary data container for the scraper. cIt inherits from scrapy.Item and defines the structure of the extracted data from Github.

2.1.1 Fields

- **url**: Stores the absolute URL of the repository. Using TakeFirst(), data is saved as a string rather than a list.
- **about**: Stores the repository description. In the case of there being no description, fallback priority logic is utilized so that the repository name is captured instead.
- **languages**: Captures the primary programming language captured by GitHub.
- **last_updated**: Stores the timestamp of the most recent commit or change.
- **commits**: A placeholder field used to track the volume of activity within the repository.

2.1.2 Methods

While scrapy.Item is primarily a data structure, the following processing logic was implemented via Processors:

- MapCompose(Clean_data): An input processor method that applies a custom cleaning function to every field, removing unwanted whitespace and newline characters.
- TakeFirst(): An output processor method that extracts the first valid element from the scrapped data list, ensuring the XML and Database receive clean, singular values.

2.1.3 Functions

clean_data(value): A custom helper function defined within the item script. It checks if a value exists and applies the .strip() method to ensure “clean” text entry. This prevents “dirty data” (like extra spaces) from cluttering the final XML output.

2.2 Class: GithubSpider

This class inherits from scrapy.Spider and contains the logic for navigating GitHub's structure and parsing the HTML response.

2.2.1 Fields

- **name:** Set to "github". This is the unique identifier used to run the spider via the command line (scrapy crawl github).
- **allowed_domains:** Restricted to [["github.com"](https://github.com)] to ensure the spider does not accidentally wander onto external websites.
- **start_urls:** Contains the initial URL of the target GitHub profile's repository page.

2.2.2 Methods

- parse(self, response): This is the primary callback method. It identifies the container for each repository using CSS selectors and loops through them to initialise the ItemLoader.
- response.urljoin(): A built-in Scrapy method used within the loop to merge the relative href paths (eg. /user/repo) with base GitHub domain to create absolute URLs.

2.2.3 Functions

- **CSS Selection:** The spider uses specialized CSS selectors (e.g. `li[itemprop="owns"]`) to pinpoint data within GitHub's complex DOM structure.
- **Fallback Logic:** Within the parsing function, a dual-selector approach was implemented for the "About" field. The spider first attempts to scrape the `p[itemprop="description"]`. If it returns null, it automatically switches to the `h3::a` text, which is the repository name. This is to ensure no empty fields are exported to the XML.

Chapter 3 Results

3.1 Result 1

The spider successfully crawled the GitHub profile and extracted 9 repository records. The data was exported to a structured XML format.

```
(venv) PS C:\Users\wtentz\Downloads\Midterm ACP\midterm_acp> scrapy crawl github -o output.xml
2026-04-16 21:05:15 [scrapy.core.scrapers] DEBUG: Scraped from <200 https://github.com/wangyalal/Midterm-IT>
{'about': 'Midterm-IT',
 'commits': '0',
 'languages': 'HTML',
 'last_updated': '2026-04-12T11:28:44Z',
 'url': 'https://github.com/wangyalal/Midterm-IT'}
2026-04-16 21:05:17 [scrapy.core.engine] DEBUG: Crawled (200) <GET https://github.com/wangyalal/Group1> (referer: https://github.com/wangyalal?tab=repositories)
2026-04-16 21:05:17 [scrapy.core.scrapers] DEBUG: Scraped from <200 https://github.com/wangyalal/Group1>
{'about': 'Group1',
 'commits': '0',
 'last_updated': '2026-04-16T11:23:14Z',
 'url': 'https://github.com/wangyalal/Group1'}
2026-04-16 21:05:17 [scrapy.core.engine] INFO: Closing spider (finished)
2026-04-16 21:05:17 [scrapy.extensions.feedexport] INFO: Stored xml feed (9 items) in: output.xml
2026-04-16 21:05:17 [scrapy.statscollectors] INFO: Dumping Scrapy stats:
{'downloader/request_bytes': 6679,
 'downloader/request_count': 10,
 'downloader/request_method_count/GET': 10,
 'downloader/response_bytes': 512783,
 'downloader/response_count': 10,
 'downloader/response_status_count/200': 10,
 'elapsed_time_seconds': 11.770435,
 'feedexport/success_count/FileFeedStorage': 1,
 'finish_reason': 'finished',
 'finish_time': datetime.datetime(2026, 4, 16, 13, 5, 17, 235053, tzinfo=datetime.timezone.utc),
 'httpcompression/response_bytes': 2418927,
 'httpcompression/response_count': 10,
 'item_scraped_count': 9,
 'items_per_minute': 49.09090909090909,
 'log_count/DEBUG': 19,
 'log_count/INFO': 3,
 'request_depth_max': 1,
 'response_received_count': 10,
 'responses_per_minute': 54.54545454545455,
 'scheduler/dequeued': 10,
 'scheduler/dequeued/memory': 10,
 'scheduler/enqueued': 10,
 'scheduler/enqueued/memory': 10,
 'start_time': datetime.datetime(2026, 4, 16, 13, 5, 5, 464618, tzinfo=datetime.timezone.utc)}
2026-04-16 21:05:17 [scrapy.core.engine] INFO: Spider closed (finished)
(venv) PS C:\Users\wtentz\Downloads\Midterm ACP\midterm_acp>
```

3.2 Result 2

All relative paths were successfully converted to absolute URLs, and the fallback logic ensured no null entries were recorded.

```
1  <?xml version="1.0" encoding="utf-8"?>
2  <items>
3    <item>
4      <url>https://github.com/wangyalal/Article-text-file</url>
5      <about>Text extraction from Kuensel's Dzongkha new articles for Dzongkha AI</about>
6      <last_updated>2025-04-17T05:13:42Z</last_updated>
7      <languages>Python</languages>
8      <commits>0</commits>
9    </item>
10   <item>
11     <url>https://github.com/wangyalal/Dzongkha-Text-Extraction</url>
12     <about>Text extraction from Dzongkha textbooks</about>
13     <last_updated>2025-04-17T06:04:11Z</last_updated>
14     <languages>Jupyter Notebook</languages>
15     <commits>0</commits>
16   </item>
17   <item>
18     <url>https://github.com/wangyalal/Learning</url>
19     <about>Learning</about>
20     <last_updated>2025-04-17T06:15:15Z</last_updated>
21     <commits>0</commits>
22   </item>
23   <item>
24     <url>https://github.com/wangyalal/Information-And-Technology-Assignments</url>
25     <about>Information-And-Technology-Assignments</about>
26     <last_updated>2026-03-09T03:39:41Z</last_updated>
27     <languages>HTML</languages>
28     <commits>0</commits>
29   </item>
30   <item>
31     <url>https://github.com/wangyalal/W4-Assignment</url>
32     <about>W4-Assignment</about>
33     <last_updated>2026-03-22T14:46:15Z</last_updated>
34     <languages>HTML</languages>
35     <commits>0</commits>
36   </item>
```

```
37   <item>
38     <url>https://github.com/wangyalal/W5-Assignment</url>
39     <about>W5-Assignment</about>
40     <last_updated>2026-03-31T13:14:10Z</last_updated>
41     <languages>HTML</languages>
42     <commits>0</commits>
43   </item>
44   <item>
45     <url>https://github.com/wangyalal/W6-Assignment</url>
46     <about>W6-Assignment</about>
47     <last_updated>2026-04-07T14:13:15Z</last_updated>
48     <languages>HTML</languages>
49     <commits>0</commits>
50   </item>
51   <item>
52     <url>https://github.com/wangyalal/Midterm-IT</url>
53     <about>Midterm-IT</about>
54     <last_updated>2026-04-12T11:28:44Z</last_updated>
55     <languages>HTML</languages>
56     <commits>0</commits>
57   </item>
58   <item>
59     <url>https://github.com/wangyalal/Group1</url>
60     <about>Group1</about>
61     <last_updated>2026-04-16T11:23:14Z</last_updated>
62     <commits>0</commits>
63   </item>
64 </items>
```

Chapter 4 Conclusions

The objective of this project was to design and implement a web scraping system using the Scrapy framework. Through the development of the github spider, I was able to successfully automate the extraction of repository metadata, transforming unstructured HTML from a personal profile into a clean, structured XML dataset.

4.1 Key Achievements

- Technical Proficiency: I successfully implemented advanced Scrapy features, including Item Loaders and Processors, which ensured high data integrity and consistency.
- Dynamic Logic: By integrating fallback mechanisms for descriptions and synchronous URL joining, the program proved capable of handling real-world web data inconsistencies.
- Data Sanitization: The use of custom cleaning functions ensured that the final output was free of “noise” and ready for professional use.

4.2 Challenges and Learning

- One of the primary challenges was handling GitHub’s relative URL paths. Resolving this using `response.urljoin()` provided a deeper understanding of how web crawlers navigate site hierarchies.
- Implementing the fallback logic required a deeper understanding of Scrapy’s Item Loader priority system. This helped me in understanding how to bridge the gap between “messy” real world websites and the “clean” structured data requirement.

4.3 Final Reflection

This project demonstrated that web scraping is not just about grabbing text from the desired websites, but about engineering a pipeline that produces clean and accessible data as well. It was a very educational project to work on, and has helped me dive deeper into the world of web scraping.